



School of Information Technology and Engineering
Security of Operating System

SIS 3

Prepare and install packages and applications, work with firewall.

“Secure SWIFT Logical Terminal Infrastructure”

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Target: Prepare and install packages and applications, work with firewall
Infrastructure context: 2 machines:

- **swift-terminal** (main SWIFT logical terminal: Nginx + backend + DB + LT storage)
- **swift-support** (central logs + backups + monitoring)

1) Needed packages & services

1.1 Package/Service table

Machine 1: swift-terminal (Fedora)

Component	Packages / Install tool	Service (systemd)	Purpose
Firewall	firewalld	firewalld	Control inbound ports (443/22 only)
Web reverse proxy	nginx	nginx	HTTPS entrypoint, serve UI, proxy /api
Backend runtime	python3, python3-pip, python3-venv	swift-backend (custom)	FastAPI app runs as swiftapi
Backend libs	fastapi, uvicorn, gunicorn, psycopg2-binary (pip)	(used by swift-backend)	API logic + DB driver
Database	postgresql-server, postgresql	postgresql	Metadata DB (local only)
Logging forwarder	rsyslog	rsyslog	Forward logs to swift-support

Package 1 — firewalld (Method 1: Fedora repo)

Install:

```
sudo dnf install -y firewalld
```

```
swift-terminal@fedora:~$ sudo dnf install firewalld
Updating and loading repositories:
Repositories loaded.
Package "firewalld-2.3.1-5.fc43.noarch" is already installed.

Nothing to do.
swift-terminal@fedora:~$
```

Enable + Start:

sudo systemctl enable --now firewalld

```
swift-terminal@fedora:~$ sudo systemctl enable --now firewalld
```

Configure rules:

Allow SSH + HTTP (add HTTPS only if you later configure it):

sudo firewall-cmd --permanent --add-service=ssh

sudo firewall-cmd --permanent --add-service=http

sudo firewall-cmd --reload

sudo firewall-cmd --list-all

```
swift-terminal@fedora:~$ sudo firewall-cmd --permanent --add-service=ssh
Warning: ALREADY_ENABLED: ssh
success
swift-terminal@fedora:~$ sudo firewall-cmd --permanent --add-service=http
bash: sudo: command not found...
swift-terminal@fedora:~$ sudo firewall-cmd --permanent --add-service=http
success
swift-terminal@fedora:~$ sudo firewall-cmd --reload
success
swift-terminal@fedora:~$ sudo firewall-cmd --list-all
FedoraWorkstation (default, active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: dhcpv6-client http samba-client ssh
  ports: 1025-65535/udp 1025-65535/tcp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
```

Package 2 — nginx (Fedora repo)

Install

sudo dnf install -y nginx

```
swift-terminal@fedora:~$ sudo dnf install nginx
Updating and loading repositories:
Repositories loaded.
Package "nginx-2:1.28.2-1.fc43.x86_64" is already installed.

Nothing to do.
```

Enable service

sudo systemctl enable --now nginx

systemctl status nginx

```
swift-terminal@fedora:~$ sudo dnf install nginx
Updating and loading repositories:
Repositories loaded.
Package "nginx-2:1.28.2-1.fc43.x86_64" is already installed.

Nothing to do.
swift-terminal@fedora:~$ sudo systemctl enable --now nginx
Created symlink '/etc/systemd/system/multi-user.target.wants/nginx.service' → '/usr/lib/systemd/system/nginx.service'.
swift-terminal@fedora:~$ systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; >
   Drop-In: /usr/lib/systemd/system/service.d
            └─10-timeout-abort.conf
   Active: active (running) since Fri 2026-03-06 02:37:29 +05; 20s>
 Invocation: e1972bb588594c8ea3741fe06a512c88
    Process: 4127 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=e>
    Process: 4130 ExecStartPre=/usr/sbin/nginx -t (code=exited, stat>
    Process: 4134 ExecStart=/usr/sbin/nginx (code=exited, status=0/S>
 Main PID: 4135 (nginx)
    Tasks: 2 (limit: 4784)
  Memory: 4.4M (peak: 4.6M)
     CPU: 59ms
   CGroup: /system.slice/nginx.service
            └─4135 "nginx: master process /usr/sbin/nginx"
              └─4137 "nginx: worker process"
```

Package 3 — PostgreSQL (dnf method)

Install

```
sudo dnf install postgresql-server postgresql -y
```

```
swift-terminal@fedora:~$ sudo dnf install postgresql-server postgresql
Updating and loading repositories:
Repositories loaded.
Package "postgresql-server-18.1-1.fc43.x86_64" is already installed.
Package "postgresql-18.1-1.fc43.x86_64" is already installed.

Nothing to do.
```

Initialize database

```
sudo postgresql-setup --initdb
```

```
swift-terminal@fedora:~$ sudo postgresql-setup --initdb
* Initializing database in '/var/lib/pgsql/data'
* Initialized, logs are in /var/lib/pgsql/initdb_postgresql.log
```

Enable service

```
sudo systemctl enable --now postgresql
```

```
systemctl status postgresql
```

```
swift-terminal@fedora:~$ sudo postgresql-setup --initdb
* Initializing database in '/var/lib/pgsql/data'
* Initialized, logs are in /var/lib/pgsql/initdb_postgresql.log
swift-terminal@fedora:~$ sudo systemctl enable --now postgresql
Created symlink '/etc/systemd/system/multi-user.target.wants/postgresql.service' → '/usr/lib/systemd/system/postgresql.service'.
swift-terminal@fedora:~$ systemctl status postgresql
● postgresql.service - PostgreSQL database server
   Loaded: loaded (/usr/lib/systemd/system/postgresql.service; enabled; vendor preset: enabled)
   Drop-In: /usr/lib/systemd/system/service.d
            └─10-timeout-abort.conf
   Active: active (running) since Fri 2026-03-06 02:41:08 +05; 11s ago
 Invocation: f20873a23777486ca2eec20c54933ae1
  Process: 4504 ExecStartPre=/usr/libexec/postgresql-check-db-dir (code=0, status=0/SUCCESS)
 Main PID: 4509 (postgres)
   Tasks: 10 (limit: 4784)
  Memory: 21.8M (peak: 22M)
    CPU: 44ms
   CGroup: /system.slice/postgresql.service
            └─4509 /usr/bin/postgres -D /var/lib/pgsql/data
               4511 "postgres: logger "
               4512 "postgres: io worker 0"
               4513 "postgres: io worker 1"
               4514 "postgres: io worker 2"
               4515 "postgres: checkpointer "
               4516 "postgres: background writer "
```

Package 4 — Python (dnf method)

Install

`sudo dnf install python3 python3-pip python3-venv -y`

```
swift-terminal@fedora:~$ sudo dnf install python3 python3-pip python-venv
Updating and loading repositories:
Repositories loaded.
Failed to resolve the transaction:
Package "python3-3.14.0-1.fc43.x86_64" is already installed.
No match for argument: python-venv
You can try to add to command line:
  --skip-unavailable to skip unavailable packages
swift-terminal@fedora:~$ python3 --version
Python 3.14.0
```

Package 5 — Backend API (pip installation)

Check user and make directory:

```
swift-terminal@fedora:~$ id swiftapi
uid=972(swiftapi) gid=969(swiftapi) groups=969(swiftapi)
swift-terminal@fedora:~$ sudo mkdir -p /opt/swift-backend
swift-terminal@fedora:~$ sudo chown swiftapi:swifapi /opt/swift-backend
chown: invalid group: 'swiftapi:swifapi'
swift-terminal@fedora:~$ sudo chown swiftapi:swiftapi /opt/swift-backend
swift-terminal@fedora:~$ ls -ld /opt/swift-backend
ls: cannot access '/opt/swift-backend': No such file or directory
swift-terminal@fedora:~$ ls -ld /opt/swift-backend
drwxr-xr-x. 1 swiftapi swiftapi 0 Mar  6 02:45 /opt/swift-backend
swift-terminal@fedora:~$
```

Create the Python virtual environment:

`sudo -u swiftapi python3 -m venv /opt/swift-backend/venv`

```

swift-terminal@fedora:~$ sudo -u swiftapi python3 -m venv /opt/swift-backend/venv
ls swift-terminal@fedora:~$ ls /opt/swift-backend
venv
swift-terminal@fedora:~$ S

```

Install backend libraries:

1. astAPI
2. Uvicorn
3. Gunicorn
4. PostgreSQL driver

```

swift-terminal@fedora:~$ sudo -u swiftapi /opt/swift-backend/venv/bin
/pip install fastapi uvicorn gunicorn psycpg2-binary
WARNING: The directory '/home/swiftapi/.cache/pip' or its parent dire
ctory is not owned or is not writable by the current user. The cache
has been disabled. Check the permissions and owner of that directory.
If executing pip with sudo, you should use sudo's -H flag.
Collecting fastapi
  Downloading fastapi-0.135.1-py3-none-any.whl.metadata (30 kB)
Collecting uvicorn
  Downloading uvicorn-0.41.0-py3-none-any.whl.metadata (6.7 kB)
Collecting gunicorn
  Downloading gunicorn-25.1.0-py3-none-any.whl.metadata (5.5 kB)
Collecting psycpg2-binary
  Downloading psycpg2_binary-2.9.11-cp314-cp314-manylinux2014_x86_64
.manylinux_2_17_x86_64.whl.metadata (4.9 kB)
Collecting starlette>=0.46.0 (from fastapi)
  Downloading starlette-0.52.1-py3-none-any.whl.metadata (6.3 kB)
Collecting pydantic>=2.7.0 (from fastapi)
  Downloading pydantic-2.12.5-py3-none-any.whl.metadata (90 kB)
Collecting typing-extensions>=4.8.0 (from fastapi)
  Downloading typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3
kB)
Collecting typing-inspection>=0.4.2 (from fastapi)
  Downloading typing_inspection-0.4.2-py3-none-any.whl.metadata (2.6
kB)

```

```
Downloading pydantic_core-2.41.5-cp314-cp314-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (2.1 MB)
2.1/2.1 MB 3.2 MB/s eta 0:00:00
Downloading annotated_types-0.7.0-py3-none-any.whl (13 kB)
Downloading starlette-0.52.1-py3-none-any.whl (74 kB)
Downloading anyio-4.12.1-py3-none-any.whl (113 kB)
Downloading idna-3.11-py3-none-any.whl (71 kB)
Downloading typing_extensions-4.15.0-py3-none-any.whl (44 kB)
Downloading typing_inspection-0.4.2-py3-none-any.whl (14 kB)
Downloading packaging-26.0-py3-none-any.whl (74 kB)
Installing collected packages: typing-extensions, psycpg2-binary, packaging, idna, h11, click, annotated-types, annotated-doc, uvicorn, typing-inspection, pydantic-core, gunicorn, anyio, starlette, pydantic, fastapi
Successfully installed annotated-doc-0.0.4 annotated-types-0.7.0 anyio-4.12.1 click-8.3.1 fastapi-0.135.1 gunicorn-25.1.0 h11-0.16.0 idna-3.11 packaging-26.0 psycpg2-binary-2.9.11 pydantic-2.12.5 pydantic-core-2.41.5 starlette-0.52.1 typing-extensions-4.15.0 typing-inspection-0.4.2 uvicorn-0.41.0
```

Create the backend program

`sudo nano /opt/swift-backend/main.py`

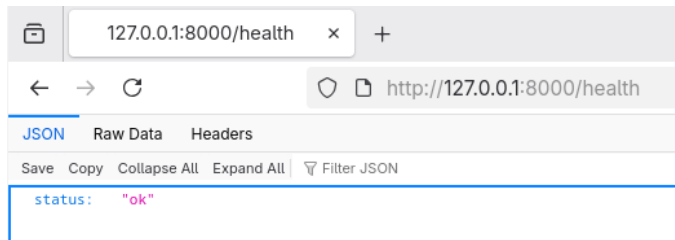
```
swift-terminal@fedora:~ — sudo nano /opt/swift-backend/main.py
GNU nano 8.5 /opt/swift-backend/main.py
from fastapi import FastAPI

app = FastAPI()

@app.get("/health")
def health():
    return {"status": "ok"}
```

Test the program manually:

```
swift-terminal@fedora:/opt/swift-backend$ sudo -u swiftapi bash -c "cd /opt/swift-backend && ./venv/bin/uvicorn main:app --host 127.0.0.1 --port 8000"
INFO: Started server process [9685]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
```

Package 6 — Custom backend systemd service

Create service

```
sudo nano /etc/systemd/system/swift-backend.service
```

A screenshot of a terminal window titled 'swift-terminal@fedora:/opt/swift-backend'. The terminal shows the contents of the file `/etc/systemd/system/swift-backend.service` being edited with nano. The file content is as follows:

```
GNU nano 8.5 /etc/systemd/system/swift-backend.service
[Unit]
Description=Swift Backend API
After=network.target

[Service]
User=swiftapi
WorkingDirectory=/opt/swift-backend
Environment="PATH=/opt/swift-backend/venv/bin"
ExecStart=/opt/swift-backend/venv/bin/uvicorn main:app --host 127.0.0.1
Restart=always

[Install]
WantedBy=multi-user.target
```

Reload

```
sudo systemctl daemon-reload
```

Enable service

```
sudo systemctl enable --now swift-backend
```

```
systemctl status swift-backend
```

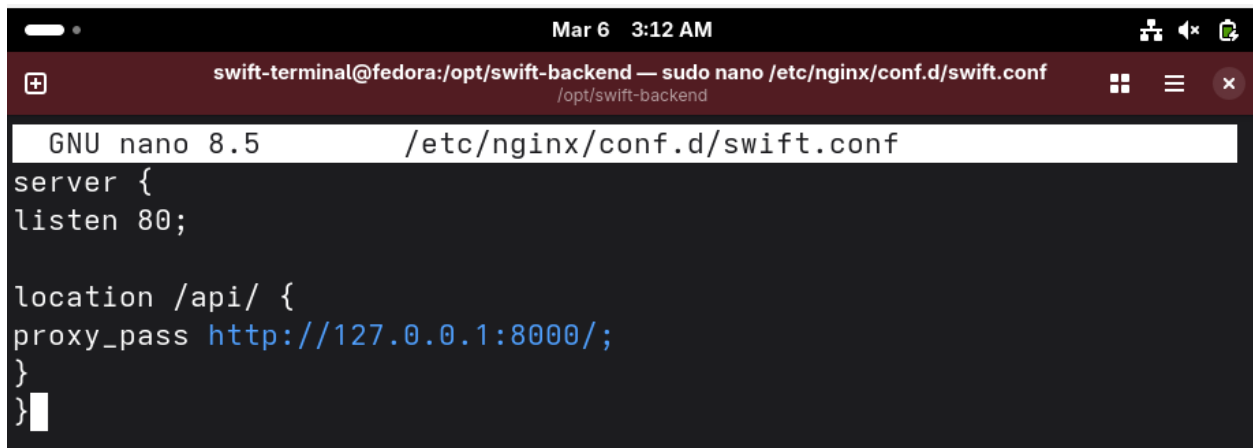
```

swift-terminal@fedora:/opt/swift-backend$ sudo systemctl daemon-reload
swift-terminal@fedora:/opt/swift-backend$ sudo systemctl enable --now swift-backend
Failed to enable unit: Unit swift-backend.service does not exist
swift-terminal@fedora:/opt/swift-backend$ sudo systemctl enable --now swift-backend
Created symlink '/etc/systemd/system/multi-user.target.wants/swift-backend.service' → '/etc/systemd/system/swift-backend.service'.
swift-terminal@fedora:/opt/swift-backend$

```

Configure nginx reverse proxy:

`sudo nano /etc/nginx/conf.d/swift.conf`



The screenshot shows a terminal window titled "swift-terminal@fedora:/opt/swift-backend — sudo nano /etc/nginx/conf.d/swift.conf". The terminal displays the contents of the file `/etc/nginx/conf.d/swift.conf` being edited with GNU nano 8.5. The configuration includes a `server` block listening on port 80, and a `location /api/` block that proxies requests to `http://127.0.0.1:8000/`. The cursor is at the end of the file.

```

GNU nano 8.5 /etc/nginx/conf.d/swift.conf
server {
listen 80;

location /api/ {
proxy_pass http://127.0.0.1:8000/;
}
}

```

Allow nginx to connect to the backend

```

swift-terminal@fedora:/opt/swift-backend$ sudo setsebool -P httpd_can_network_connect on
swift-terminal@fedora:/opt/swift-backend$ getsebool httpd_can_network_connect
httpd_can_network_connect --> on

```

Restart

`sudo systemctl restart nginx`

`nginx -t`

```

swift-terminal@fedora:/opt/swift-backend$ sudo systemctl restart nginx
x

```

```

swift-terminal@fedora:/opt/swift-backend$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful

```

Machine 2: swift-support (Fedora)

Component	Packages / Install tool	Service (systemd)	Purpose
Firewall	firewalld	firewalld	Allow only SSH + syslog + admin tools
Central log receiver	rsyslog	rsyslog	Receive logs from swift-terminal
Backups	rsync, tar	(cron/timer later)	Store backups under /backup/swift
Monitoring (repo install)	grafana (via added repo)	grafana-server	Basic monitoring UI
Monitoring (archive install)	node_exporter (tar.gz)	node_exporter (custom)	Host metrics exporter

Package 1 — rsyslog (dnf method)

Install

```
sudo dnf install rsyslog -y
```

```

swift-support@fedora:~/Desktop$ sudo dnf install rsyslog
Updating and loading repositories:
Repositories loaded.
Package "rsyslog-8.2508.0-1.fc43.x86_64" is already installed.

Nothing to do.

```

Enable

`sudo systemctl enable --now rsyslog`

`systemctl status rsyslog`

```
swift-support@fedora:~/Desktop$ sudo systemctl enable --now rsyslog
swift-support@fedora:~/Desktop$ systemctl status rsyslog
● rsyslog.service - System Logging Service
   Loaded: loaded (/usr/lib/systemd/system/rsyslog.service; enable>
   Drop-In: /usr/lib/systemd/system/service.d
            └─10-timeout-abort.conf
   Active: active (running) since Fri 2026-03-06 03:18:20 +05; 4mi>
  Invocation: 43aee6af61aa4b1a969585d22131a8c7
     Docs: man:rsyslogd(8)
           https://www.rsyslog.com/doc/
   Main PID: 891 (rsyslogd)
      Tasks: 3 (limit: 4784)
     Memory: 9.1M (peak: 9.3M)
        CPU: 846ms
    CGroup: /system.slice/rsyslog.service
            └─891 /usr/sbin/rsyslogd -n

Mar 06 03:18:21 fedora rsyslogd[891]: imjournal: journal files chang>
```

Package 2 — rsync (dnf method)

`sudo dnf install rsync -y`

```
swift-support@fedora:~/Desktop$ sudo dnf install rsync -y
Updating and loading repositories:
Repositories loaded.
Package "rsync-3.4.1-4.fc43.x86_64" is already installed.

Nothing to do.
swift-support@fedora:~/Desktop$ rsync --version
rsync version 3.4.1 protocol version 32
Copyright (C) 1996-2025 by Andrew Tridgell, Wayne Davison, and others
.
```

Package 3 — tar (dnf method)

```
sudo dnf install tar -y
```

```
swift-support@fedora:~/Desktop$ sudo dnf install tar
Updating and loading repositories:
Repositories loaded.
Package "tar-2:1.35-6.fc43.x86_64" is already installed.

Nothing to do.
swift-support@fedora:~/Desktop$ tar --version
tar (GNU tar) 1.35
Copyright (C) 2023 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <https://gnu.org/licenses/gpl.html>.
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

Written by John Gilmore and Jay Fenlason.
```

Package 4 — Grafana (NEW REPOSITORY METHOD)

```
sudo nano /etc/yum.repos.d/grafana.repo
```

```
swift-support@fedora:~/Desktop — sudo nano /etc/yum.repos.d/grafana.repo
~/Desktop
GNU nano 8.5 /etc/yum.repos.d/grafana.repo
[grafana]
name=grafana
baseurl=https://rpm.grafana.com
enabled=1
gpgcheck=1
gpgkey=https://rpm.grafana.com/gpg.key
```

Install Grafana

```
sudo dnf makecache
```

```
sudo dnf install grafana -y
```

```

swift-support@fedora:~/Desktop$ sudo dnf makecache
Updating and loading repositories:
  grafana          100% | 217.4 KiB/s | 343.5 KiB | 00m02s
Repositories loaded.
Metadata cache created.
swift-support@fedora:~/Desktop$ sudo dnf install grafana
Updating and loading repositories:
Repositories loaded.
Package           Arch   Version           Repository        Size
Installing:
  grafana          x86_64 12.4.0-1          grafana           690.5 MiB

Transaction Summary:
  Installing:          1 package

Total size of inbound packages is 188 MiB. Need to download 188 MiB.
After this operation, 691 MiB extra will be used (install 691 MiB, remove 0 B).
Is this ok [y/N]: y
[1/1] grafana-0:12.4.0-1.x86_64 100% | 1.5 MiB/s | 187.6 MiB | 02m06s
-----
[1/1] Total                  100% | 1.5 MiB/s | 187.6 MiB | 02m06s
[1/2] https://rpm.grafana.co ???% | 0.0 B/s | 0.0 B | 00m00s
[1/2] https://rpm.grafana.co ???% | 0.0 B/s | 0.0 B | 00m00s
[1/2] https://rpm.grafana.co ???% | 0.0 B/s | 0.0 B | 00m00s

```

Enable service

`sudo systemctl enable --now grafana-server`

`systemctl status grafana-server`

```

swift-support@fedora:~/Desktop$ sudo systemctl enable --now grafana-server
[sudo] password for swift-support:
Created symlink '/etc/systemd/system/multi-user.target.wants/grafana-server.service' → '/usr/lib/systemd/system/grafana-server.service'.
swift-support@fedora:~/Desktop$ systemctl status grafana-server
● grafana-server.service - Grafana instance
   Loaded: loaded (/usr/lib/systemd/system/grafana-server.service;
   Drop-In: /usr/lib/systemd/system/service.d
           └─10-timeout-abort.conf
   Active: active (running) since Fri 2026-03-06 03:34:27 +05; 17s
   Invocation: 73e244a9e8a144c58d9ac6273819449d
   Docs: http://docs.grafana.org
   Main PID: 4979 (grafana)
   Tasks: 15 (limit: 4784)
   Memory: 291M (peak: 291.3M)

```

Package 5 — Node Exporter (ARCHIVE METHOD)

Download

```
cd /tmp
```

```
curl -LO
```

https://github.com/prometheus/node_exporter/releases/download/v1.8.2/node_exporter-1.8.2.linux-amd64.tar.gz

```
swift-support@fedora:/tmp$ cd /tmp
swift-support@fedora:/tmp$ curl -LO https://github.com/prometheus/node_exporter/releases/download/v1.8.2/node_exporter-1.8.2.linux-amd64.tar.gz
  % Total    % Received % Xferd  Average Speed   Time    Time     Time
  e  Current          Dload  Upload   Total   Spent    Lef
t  Speed
  0     0    0     0    0     0      0      0  --:--:--  --:--:--  --:--
  0     0    0     0    0     0      0      0  --:--:--  --:--:--  --:--
  0     0    0     0    0     0      0      0  --:--:--  --:--:--  --:--
:--    0
  0 10.1M    0 51394    0     0 48243    0  0:03:41  0:00:01  0:03
35 10.1M   35 3709k    0     0 1772k    0  0:00:05  0:00:02  0:00
100 10.1M  100 10.1M    0     0 4144k    0  0:00:02  0:00:02  --:--
:-- 7155k
swift-support@fedora:/tmp$
```

Extract:

```
tar -xzf node_exporter-1.8.2.linux-amd64.tar.gz
```

```
swift-support@fedora:/tmp$ tar -xzf node_exporter-1.8.2.linux-amd64.tar.gz
```

Move the binary to system folder:

```
swift-support@fedora:/tmp$ sudo mv node_exporter-1.8.2.linux-amd64/node_exporter /usr/local/bin/
[sudo] password for swift-support:
```

Create the service:

```
swift-support@fedora:/tmp — sudo nano /etc/systemd/system/node_exporter.service
GNU nano 8.5 /etc/systemd/system/node_exporter.service
[Unit]
Description=Node Exporter

[Service]
ExecStart=/usr/local/bin/node_exporter
Restart=always

[Install]
WantedBy=multi-user.target
```

Start the service:

```
swift-support@fedora:/tmp$ ls -Z /usr/local/bin/node_exporter
unconfined_u:object_r:bin_t:s0 /usr/local/bin/node_exporter
swift-support@fedora:/tmp$ sudo systemctl daemon-reload
sudo systemctl restart node_exporter
systemctl status node_exporter --no-pager
● node_exporter.service - Node Exporter
   Loaded: loaded (/etc/systemd/system/node_exporter.service; enabled; preset: disabled)
   Drop-In: /usr/lib/systemd/system/service.d
            └─10-timeout-abort.conf
   Active: active (running) since Fri 2026-03-06 04:02:19 +05; 169ms ago
     Invocation: 026e9b745abb48d3ba1feb88c5409df0
       Main PID: 7054 (node_exporter)
         Tasks: 3 (limit: 4784)
        Memory: 2.1M (peak: 2.1M)
           CPU: 16ms
```

Firewall configuration

```
sudo firewall-cmd --permanent --add-port=3000/tcp
sudo firewall-cmd --permanent --add-port=9100/tcp
sudo firewall-cmd --reload
```



```
swift-support@fedora:/$ sudo firewall-cmd --permanent --add-port=3000/tcp
[sudo] password for swift-support:
Warning: ALREADY_ENABLED: 3000:tcp
success
swift-support@fedora:/$ sudo firewall-cmd --permanent --add-port=9100/tcp
Warning: ALREADY_ENABLED: 9100:tcp
success
swift-support@fedora:/$ sudo firewall-cmd --reload
success
swift-support@fedora:/$
```

Smoke Test:

1. Backend test:

On swift-terminal -

curl http://127.0.0.1:8000/health

```
swift-terminal@fedora:/opt/swift-backend$ curl http://127.0.0.1:8000/health
{"status":"ok"}swift-terminal@fedora:/opt/swift-backend$
```

2. Nginx proxy test

Run on swift-terminal

curl http://127.0.0.1/api/health

```
swift-terminal@fedora:/opt/swift-backend$ curl -i http://127.0.0.1/api/health
HTTP/1.1 200 OK
Server: nginx/1.28.2
Date: Thu, 05 Mar 2026 23:46:58 GMT
Content-Type: application/json
Content-Length: 15
Connection: keep-alive
```

3. Grafana test

on swift-support

curl http://localhost:3000

```
swift-support@fedora:/$ curl http://localhost:3000
<a href="/login">Found</a>.

swift-support@fedora:/$
```

4. Node Exporter test

on swift-support

curl http://127.0.0.1:9100/metrics | head

```
swift-support@fedora:/$ curl http://127.0.0.1:9100/metrics | head
% Total    % Received % Xferd  Average Speed   Time    Time     Tim
e  Current                                  Dload  Upload  Total  Spent  Lef
t  Speed
0      0    0     0     0     0     0     0  --:--:-- --:--:-- --:--:
:--      0# HELP go_gc_duration_seconds A summary of the pause duratio
n of garbage collection cycles.
# TYPE go_gc_duration_seconds summary
go_gc_duration_seconds{quantile="0"} 1.8006e-05
go_gc_duration_seconds{quantile="0.25"} 4.377e-05
1go_gc_duration_seconds{quantile="0.5"} 8.8051e-05
0go_gc_duration_seconds{quantile="0.75"} 0.000147446
0go_gc_duration_seconds{quantile="1"} 0.000370715
  go_gc_duration_seconds_sum 0.00250383
7go_gc_duration_seconds_count 24
1# HELP go_goroutines Number of goroutines that currently exist.
114    0 71114    0    0 1525k    0 --:--:-- --:--:-- --:--:-- 15
43k
curl: Failed writing body
```

5. Firewall blocking test (VERY IMPORTANT)

Run on swift-support; Try to access PostgreSQL on swift-terminal.

First find swift-terminal IP:

```

1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN g
roup default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel
state UP group default qlen 1000
    link/ether 08:00:27:db:16:20 brd ff:ff:ff:ff:ff:ff
    altname enx080027db1620
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixrou
te enp0s3
        valid_lft 84421sec preferred_lft 84421sec
    inet6 fd17:625c:f037:2:7a3c:7706:b96:9ece/64 scope global dynamic
noprefixroute
        valid_lft 86295sec preferred_lft 14295sec
    inet6 fe80::4723:32e9:554d:9877/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

```

Then on swift-support run:

```

swift-support@fedora:/$ nc -zv 10.0.2.15 5432
Ncat: Version 7.92 ( https://nmap.org/ncat )
Ncat: Connection refused.
swift-support@fedora:/$

```

Conclusion

In this task, the system designed in SIS1 and configured in SIS2 was extended by installing and configuring the required software. The services were deployed on two machines according to the architecture: swift-terminal for the main application services and swift-support for monitoring and support tools. The necessary packages were installed using different methods, including system repositories, an additional repository, and archive installation. Services such as the backend API, Nginx, Grafana, and Node Exporter were configured and started using systemd. Firewall rules were also applied to allow only required network access and block unnecessary

connections. The final tests confirmed that the services are working correctly and the system follows the design defined in SIS1 and SIS2.